

Underwater Glider

This internship is open to all Newcastle University students studying relevant programmes, including Robotics, Electronic, Mechanical, Marine or Software Engineering.

This internship is expected to be for a total of 60 hours. You will work up to 6 hours per week between February and July 2026. Working days and times are flexible.

Work will take place on campus, Newcastle Autonomous Marine Vehicle Lab, Stephenson Building.

This project aims to develop a next-generation autonomous underwater vehicle (AUV), specifically a buoyancy-driven underwater glider, designed to enhance ocean observation capabilities. Unlike conventionally propelled AUVs, the underwater glider achieves motion by adjusting its buoyancy, allowing it to ascend and descend through the water column while converting vertical movement into horizontal glide. This energy-efficient mechanism enables long-duration and wide-range missions, making it an ideal platform for sustained environmental monitoring.

We are looking for a talented student in control and electrical systems to develop novel control software and systems together with the research team. These are two potential tasks you could undertake as an intern; one focusing on hardware in circuit design and PCB development and the other one focusing on software development in Python or C++. As an intern you will join the lab to develop skills and expertise in underwater glider control system and will have the opportunity to experience the leading-edge underwater autonomous systems. For technical details, please see this website, <https://abm-tech.co.uk/>.

Key tasks:

We are looking for a technical oriented and self-motivated research intern to join our research team to together deliver the below aspects:

- Control systems: Arduino-based motion control and navigation. (C++ and Arduino language)
- Electronics: Circuit design, wiring and PCB design
- Software development: Implementation of user-friendly control software for mission planning, real-time monitoring, and data visualization. (Python)

What are we looking for?

Essential skills:

- § For candidates from Mechanical Engineering, Marine Engineering, Robotics, Control, Electrical and Electronic Engineering (EEE): Solid knowledge of circuit design and PCB development. Familiarity with electronic components, testing, and integration is preferred.

§ For candidates from Software Engineering, Computer Science (CS), Robotics:
Proficiency in programming with Python or C++. Experience in algorithm development, data processing, or control software is an advantage.

Desirable skills:

§ CAD modelling experience
§ Hands-on lab testing experience

NCL Internship: "Mind the Gap! Predicting Missing Water Meter Data using ML"

Newcastle University is offering an exciting internship opportunity for 1 student, working with the School of Engineering. The internship is being facilitated by the Jobs on Campus team in the Careers Service.

This internship is open to Stage 2 and above Newcastle University students studying Electrical and Electronics Engineering, Computer Science, or a related discipline.

This internship is expected to be for a total of 60 hours. You will work up to 5 hours per week between March and May 2026. Working days and times are flexible.

Work will take place on campus, likely on Merz Court (ISC Research Group PhD offices or the Sensors Laboratory).

In the UK, ageing water pipelines lose over three billion litres of treated water every day due to leaks that are hard to detect. Replacing entire networks is neither practical nor affordable, so water companies are turning to digital monitoring. Smart meters, combined with flow and pressure sensors, are now being deployed to track hourly consumption data, which can be interrogated to spot anomalies that may pinpoint leaks or bursts. However, these meters face a major challenge: data gaps. The UK Water Services Regulation Authority (Ofwat) requires an 85% hourly data success rate (DSR), yet current wireless technologies such as LoRaWAN and NB-IoT often deliver only 60-70%, even in good signal conditions. Closing this gap by adding more gateways, i.e., densifying the network to minimise packet losses, would cost water companies millions, involving also lengthy planning permissions and public reluctance. This project aims to address this gap using data science and machine learning to reconstruct missing water meter data, improving network reliability and leak detection without costly infrastructure expansion.

Working with a large dataset provided by Connexin, as an intern you will analyse millions of historical meter transmissions to identify patterns in household consumption and develop models to reconstruct missing readings. Methods such as recurrent neural networks (RNNs), long short-term memory (LSTM) networks, or other modern architectures (e.g. Transformer-based or GANs) may be explored, depending on progress. You will be closely supervised by Dr Oktay Cetinkaya (School of Engineering) and supported by Mr Jarod Gillespie (Head of Strategy and Delivery, Connexin), ensuring you receive ongoing guidance and feedback throughout the project. This role provides hands-on experience at the

intersection of IoT, machine learning, and sustainability, developing practical skills in data analysis and model validation while contributing to one of the UK's key infrastructure challenges.

Key tasks:

- § Preprocessing and cleaning raw smart-meter data to locate gaps (Week 1-2)
- § Training predictive models that learn from complete sequences of data (Week 3-4)
- § Testing model accuracy by deliberately removing known readings and checking how well the model predicts them, using measures like RMSE and MAE (Week 5)
- § Benchmarking against simple alternatives, such as linear interpolation (Week 5)
- § Compiling results, preparing the final report, and conducting the industrial supervisor feedback session (Week 6)

What are we looking for?

Essential skills:

- § Basic experience of Python or MATLAB
- § An interest in data analysis, IoT, distributed sensing, and machine learning/AI

Desirable skills:

- § Critical thinking and problem-solving
- § Creativity and innovation in approaching challenges
- § Strong communication and presentation skills
- § Time management and organisational ability
- § Willingness to learn new tools and methods

NCL Internship: Employability Engagement Intern (Engineering)

Degree discipline

Students studying Electrical and Electronics Engineering, Computer Science, or a related discipline are welcome to apply.

Newcastle University is offering an exciting internship opportunity for 1 student, working with the School of Engineering. The internship is being facilitated by the Jobs on Campus team in the Careers Service.

This internship is open to Stage 2 and 3 Newcastle University students from the School of Engineering.

This internship is expected to be for a total of 60 hours. You will work approximately 2-3 hours per week between December 2025 and June 2026. Working days and times are flexible.

Work will take place mainly remotely with some in person meetings and events on campus when needed. All work must be undertaken in the UK, including any work carried out remotely.

The School of Engineering is looking for an enthusiastic stage student to be the School's Employability Engagement Intern. You will collaborate with colleagues in the School and the Careers Service to support and enhance employability provision for your fellow students, while also developing your own skills and experience. You will receive training and ongoing support from the Careers Service, and your day-to-day work will be managed by a designated member of staff within the School.

Key tasks:

- § Assist with the planning, promotion, and delivery of careers related events
- § Promote relevant employability activities and opportunities to students
- § Establish connections with current students, alumni, and employers
- § Contribute to the School's employability communications, such as blogs, newsletters, or other platforms by sharing opportunities and creating content on career-related topics
- § Interview students, alumni, and employers to create engaging case studies using various forms of media
- § Engage with students to gather feedback and insights through focus groups or other methods to inform and improve employability initiatives
- § Work closely with School staff to coordinate employability-related events
- § Generate and implement new ideas to enhance the School's employability offering
- § Take ownership of your own projects and support the delivery of team initiatives
- § Actively participate in the Employability & Engagement Intern network via Teams and in person to share insights, collaborate with peers, and contribute to wider employability initiatives
- § Produce a report (in an agreed chosen format) on the learnings and successes of the internship to be used as support for future interns

What are we looking for?

Essential skills:

- § Teamwork – Ability to collaborate effectively with staff, fellow interns, and students to deliver engaging employability initiatives
- § Initiative – Proactive in identifying opportunities, suggesting ideas, and taking ownership of tasks
- § Time Management – Capable of balancing internship responsibilities with academic commitments and meeting deadlines
- § Organisation and Planning – Able to manage tasks effectively and demonstrate experience organising activities, projects, or responsibilities—whether academic, extracurricular, or personal

- § Written Communication Skills – Able to produce clear, engaging content for student communications across various platforms
- § Verbal Communication Skills - Able to build relationships with and confidently present to students during events and focus groups
- § Attention to Detail – Able to ensure accuracy and quality in written materials, event planning, and communications
- § Interest in supporting others to explore career pathways and enhance employability skills

Desirable skills:

- § Experience using social media to share information or promote activities
- § Experience managing or contributing to a social media or digital communication account
- § Creative approach to generating ideas and content

NCL Internship: Engineering Intern (Time Series Forecasting and Anomaly Detection)

Degree discipline

Stage 2 and 3 students from the School of Engineering are welcome to apply.

To be eligible to apply you must be a Newcastle University undergraduate/postgraduate student registered on a programme of study throughout the 2025-26 academic year. For full details of eligibility see the [Working on Campus website](#).

Work done as part of this role can count towards an NCL+ Award which can help you reflect on the skills you have gained and how to use them when applying for jobs in the future; to find out more visit [our webpages](#).

Newcastle University is offering an exciting internship opportunity for 1 student, working with the School of Engineering. The internship is being facilitated by the Jobs on Campus team in the Careers Service.

This internship is open to Newcastle University students who are undertaking their first undergraduate degree (or integrated Master's).

This internship is expected to be for a total of 60 hours. You will work approximately 3 hours per week between February and July 2026. Working days and times are flexible.

Work will take place mainly remotely, but you will be expected to attend weekly in-person meetings on campus in the Stephenson Building.

This internship will focus on time series forecasting and anomaly detection. Time series data are central in many engineering and scientific domains which include energy systems, industrial processes, and health monitoring. These data often show seasonal variation, noise, and sudden changes, which make forecasting and anomaly detection both difficult and meaningful. Many advanced models achieve good accuracy, yet they often act as black

boxes, which limits their trustworthiness. This project therefore aims to build methods that combine predictive accuracy.

The project itself examines classical forecasting models, machine learning methods, and deep learning approaches applied to real or synthetic time series data. The aim is to produce comparative research findings, a compact set of experimental notebooks, and a report which summarises methods, results, and open challenges. The work emphasises research thinking and scientific reporting, with the possibility of contributing to an academic publication.

As an intern you will report directly to Dr. Ekin Can Erkus (Senior Research Associate, Microsystems Group, School of Engineering) and will work in close contact with the academic supervisory team. Weekly meetings will provide progress checks, technical discussions, and planning. You will meet the industrial supervisor at least twice during the internship and a high level of support and supervision will be provided to complete the tasks.

Key tasks:

- § Clean and preprocess time series datasets and perform exploratory analysis of trends and seasonal patterns
- § Implement baseline forecasting models
- § Extend experiments to machine learning and deep learning models for forecasting and anomaly detection
- § Apply and compare anomaly detection methods which include residual rules, autoencoders, and isolation forests
- § Record results with suitable evaluation metrics and summarise findings in weekly updates
- § Prepare a final report and presentation which reflect the research outcomes

What are we looking for?

Essential skills:

- § Coding skills in Python, with ability to implement and test models
- § Understanding of time series analysis and data preprocessing
- § Knowledge of machine learning concepts, including supervised and unsupervised methods
- § Ability to write reports and communicate results clearly
- § Motivation to work independently while contributing to research discussions

Desirable skills:

- § Interest in academic writing and research
- § The internship would be most suited to those studying Engineering, Computer Science, Data Science, Artificial Intelligence, Electrical and Electronic Engineering, or Information Systems.

NCL Internship: Engineering Intern (Logic-based Machine Learning Project)

This internship is open to Newcastle University students studying BEng in EEE or ECE, BSc in Computing. The role would be most suited to those Stage 3 or above.

This internship is expected to be for a total of 360 hours. You will work up to 20 hours per week between March and July 2026. Working days and times are flexible.

Work will take place on campus in the Stephenson Building, although work can be blended. All work must be undertaken in the UK, including any work carried out remotely.

Machine learning and pervasive AI is now a hot topic in electronics and computing. The areas of application are numerous, including driverless cars, autonomous vehicles, IoT, wearable sensors. Machine intelligence and learning capabilities are becoming a commodity of any imaginable piece of electronics. Conventional machine learning is largely based on multilayer neural networks and gradient backpropagation. They are often obscure from the designer, i.e. non-explainable, and quite resource demanding, for example from energy point of view. New directions in machine learning that pursue both the issue of explainability and energy efficiency, involve using logic, in the form of Boolean propositions or even combinations of logic gates.

This project will investigate an approach to build AI systems based on the logic gate networks (LGNs), where at the stage of training a network consists of abstract logic gates, whose Boolean functions are uncommitted (each gate can potentially be NOT, AND, OR, NAND etc.). During training the gates' functions are determined using one of the feedback methods, such as Boolean gradient descent, or learning automata). The aim of the project is to build a logic gate network that can be implemented either in software or in FPGA hardware, demonstrate its capability on an example in one of the pervasive applications, such as sound or visual image recognition.

As an intern you will learn principles of AI hardware, machine learning methods, power-efficient electronic design.

The role will involve: 1) reading a set of key literature examples on logic-based machine learning models, such as logic gate networks (LGNs), 2) defining appropriate representations of data for training and testing and suitable binarization methods, 3) developing algorithms for propagating errors in such models and appropriate reinforcement rules, 4) mapping algorithms to hardware description languages such as Verilog or VHDL, 5) synthesizing circuits for FPGAs, with appropriate optimisations, 6) performing testing on datasets such as MNIST and CIFAR10, as well as some datasets from the industrial partner, 7) writing a report on the project with critical assessment of the technology and its future use.

You will be reporting to Alex Yakovlev, Professor of Computer Systems Design. Literal Labs (literal-labs.ai) Literal Labs (literal-labs.ai) will be collaborating with the University on this project and you will have regular meetings with the engineers from Literal Labs.

Key tasks:

Specific tasks will be dependent on skills of the applicants. Progression through the below responsibilities is likely over the course of the placement as the candidate becomes more experienced.

- Review of existing computing hardware, AI and machine-learning techniques
- Software or FPGA design for LGN available (github) function library
- Apply the design to a ML application
- Day to day tasks will be set during the project on Slack
- The deliverables will include a Github site with the software and hardware designs, code and progress reports

What are we looking for?**Essential skills:**

- Knowledge of digital circuit design, programming in python and/or C (C++)
- Demonstrable skills in solving some problems in logic design and programming

Desirable skills:

- Some skills/experience with machine learning, logic design, field-programmable gate arrays (FPGAs)
- Familiarity with VHDL and/or Verilog hardware description languages